explainability, fairness and non-discrimination, safety and reliability, fair competition, and privacy are part of foundational elements of Responsible AI.

Some of the elements also come from the supply side of AI systems. We see AI vendors continually pushing for faster adoption of AI solutions. I think this is greedy and not at all customer-centric approach. I can understand that vendors want to sell AI by making it easier and frictionless, but they seem to ignore inherent risks in their system as well as their solutions' ecosystem that is not in vendor's control. It is becoming more like a cart before the horse situation.

Fig. 2: A 10-step Leash System

It only benefits vendors, while enterprises often lose money and time. Because once companies start, there is no turning back. So they go deep in the rabbit-hole and keep bleeding through the nose. Cautious steps are of utmost necessity, by vendors as well as by enterprises.

However, in general, I think, it must cover three core aspects of a good solution design, i.e., you are designing the right solution, the solution has been designed right, and most importantly, you have covered all the risks.

Designing solutions as Responsible AI

While most of the companies understand the need for Responsible AI, they are unclear on how to go about it. Designing solutions as Responsible AI is still a complex and mysterious act.

While researching for my book on AI, *Keeping Your AI Under Control*, I created a 10-step approach or a framework, The Leash System. This system expertly guides solution development on the correct path, which then would help in designing the solution as Responsible AI. The ten steps are as follows:

1. Defining the problem correctly. Define what are you trying to achieve with the AI system, what is your short-term purpose and how does it align with your overall long-term purpose as well as values.

2. Establish correct levers or drivers of that problem. Without identifying the right drivers, it is highly challenging to have a correct solution in place.

3. Validate issues and its drivers. It is also crucial that by way of simulation or any other mechanism, the problem and its drivers should be validated. Not only that, you will also find it useful to identify which drivers are significant contributors and which ones are not.

4. Link AI solution with problem drivers. This advice might sound counter-intuitive, but you see, we don't solve a problem with solutions, we fix drivers of that problem with solutions, and when we set those drivers, problem automatically gets fixed.

5. Verify that fixing key problem-drivers will indeed fix your problem. Sometimes, we come across confounding variables, which appear to be drivers of a problem, but they are not. By validating thoroughly, you can ensure that you are not wasting your efforts and resources that are not fit for AI solution.

6. Understand risks. Risks can come in all sizes—small, medium, large. Regardless of that, understanding and acknowledging that it exists is the key.

7. Quantify risks. By quantifying risks, you can prioritise your approach towards risk mitigation. And, if not, you will at least be aware of potential issues.

8. Establish controls. Once you have identified and quantified risks, establish appropriate control systems, and minimise risks, or better yet eliminate those risks wherever possible.

9. The Red Team. It is not anymore just a cybersecurity paradigm. If you want a robust and trustworthy AI system, get your Red team operational.

10. Manage residual risk. And finally, despite doing all the above, if anything or any risk remains, manage that residual risk by other possible means.

When companies follow this 10-step framework, designing and deploying better AI solutions becomes entirely possible.

Designing responsibility and building trust

However, another critical question remains, "How do we design responsibility and build trust in systems?"

Let's take real-life analogy here. To trust someone in real life, we typically rely on three things:

- We rely on how someone was taught, their upbringing or education, etc.
- We test them thoroughly to ensure that they have learned it well and understand the rules of society.
- We make sure to provide them with the right operative environment to do their best.

Similarly, for AI to be trustworthy, we must ensure that:

- AI solution is trained well, which means useful data and proper training methods were used. All pos-